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Uptake studies of fluoride ions from an aqueous media using triethylamine modified watermelon rind biowastes

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Abstract
The current study utilized triethylamine modified watermelon rind biowastes in defluoridation of water. The raw adsorbent was chemically modified using thionyl chloride and triethylamine. Both raw and modified adsorbent were characterized using Thermogravimetric Analyzer (TGA), Fourier Transform Infrared (FT-IR) and Scanning Electron Microscope-Energy Dispersive X-Ray (SEM-EDX). The FT-IR results showed functional groups of hydroxyl (-OH), carbonyl (-C = O), carboxylate (-COO-) (raw adsorbent) and additional anchored amine ($-^{+}N(CH_2CH_3)_3$) groups (modified adsorbent) as important adsorption sites for fluoride ions removal. The presence of nitrogen (N) element (EDX results) confirmed the modification process. The SEM results showed a more porous adsorbent which also took part in the adsorption process. Thermal stability of the modified adsorbent was up to a temperature of 400 °C. Maximum efficiency of 85.97% was obtained at optimal values of pH (1), 50 min, 70 mg and 50 mg/L. The optimal conditions were confirmed by 3D surface plots. Adsorption of the F^- ions fitted Langmuir isotherm model and pseudo-second-order ($R^2 > 0.9$) which was more suitable in explaining the chemisorption process. Water samples from Lake Baringo water reported fluoride ion concentration of 3.59 mg/L (above WHO levels) which was reduced to 0.76 mg/L. The regeneration showed a good reusability of the adsorbent with > 50% desorption efficiency. The research findings report a potential adsorbent that can be applied in defluoridation of water to safe levels both at household and industrial scale.

Keywords Adsorption, Central composite design (CCD), Fluoride, Multivariate analysis, One analysis at a time (OVAT), Triethylamine

1 Introduction

Fluoride is a vitally and indispensable element for maintenance of animal and human health [1]. It facilitates bone mineralization if consumed at permissible concentration [2]. Although trace doses of fluoride are needed by human beings, excess fluoride has negative effects on human being and livestock [3]. According to the World Health Organization, the allowed fluoride levels in drinking water is 1.5 mg/L [4]. Low fluoride amounts in water is necessary for prevention of tooth decay and the developing proper

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bone structure [5]. Excess fluoride have severe effects on human wellbeing by the way they interact with calcium in the bones resulting to dental fluorosis and/or skeletal fluorosis (bone cancer) [6]. Other health effects are such as impotence, brittle bones, cancer, Alzheimer's disease, brain damage and thyroid disorders [7].

Besides the natural sources, discharge of the laden wastewaters from industries producing various fluoride containing products such as cosmetics, drugs, fluoridated dental product like toothpaste, glass and ceramic production, coal power plants, food, beverages, electroplating, rubber and fertilizer manufacturing industry, also contribute to surface and groundwater contamination with fluoride [1, 8]. Fluoride contamination has been reported as one of the major global problem, resulting to serious human health threat [2]. Several countries including Kenya have embraced this WHO standard as the key guideline of reducing fluoride levers in water in levels that is safe for consumption. However, as at the present, millions of folks depend on the only available fluoridated groundwater with levels exceeding WHO value. According to previous researches, this is an issue affecting both developed and developing countries [9]. In Kenya, majority of folks especially those living in Rift Valley region are facing acute fluoride problems. Even to their level of acceptance (< 1.5 mg/L), prolonged use leads to their accumulation in tissues. Therefore, to overcome the negative health effects of fluoride in water, there is need for their decontamination.

Numerous techniques, including electro dialysis, adsorption, ion-exchange, coagulation-precipitation, nanofiltration, reverse osmosis, membrane processes and biological methods [10, 11] amongst others have been used to defluoridate water. The adsorption has remained the best technique because of its greater efficacy, simplicity, environmental friendliness, easy in operation and low cost [12]. Several adsorbents, including Maize Tassels [13], Palm Kernel Shell [14], polythene [15], zirconium-based metal organic frameworks [2], Natural Pumice [6], water hyacinth [16] and seashell [17] among many others have been utilized for defluoridation.

Interaction of water and agricultural biowaste in their raw form causes leaching of soluble organic compounds leading to high chemical, biological and total oxygen demand (COD, BOD and TOD) in water leading to secondary pollution [18]. They also register low adsorption capacity. To offer a solution to the above problems, the biomass needs to be stabilized by chemical modification. The study sought to use triethylamine to modify the watermelon rind biowastes. This aimed at improving its surface porosity and efficiency compared to the reported adsorbents in the literature. The cellulose is first chlorinated and then modified as shown by Fig. 1A and B.

The modified adsorbent material possesses a positively charged amine groups that are capable of complexing fluoride ions in water without leaching soluble organic secondary pollutants in the treated water [13]. The study of chemical modification of watermelon rind biowastes via thionyl chloride activation followed by triethylamine (TEA) and the utilization of the modified adsorbent in water defluoridation remains unexplored. This forms the basis of the current study.

2 Materials and methods

2.1 Research design

The research study employed an experimental approach that entailed sample collection and pre-treatment, chemical modification, characterization, optimization of adsorption

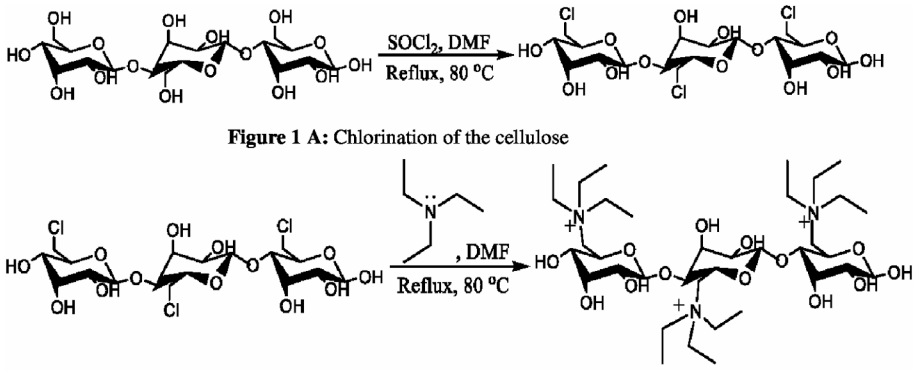

Fig. 1 **A** Chlorination of the cellulose. **B** Modification reaction scheme

parameters, equilibrium, kinetic and interactive batch experiments using synthetic waters samples.

2.2 Chemicals and reagents

Hydrochloric acid (HCl), Sodium Chloride (NaCl), Sodium Fluoride (NaF), 1,2-cyclohexanedinitrilo-tetraacetic acid (CDTA), Potassium hydroxide (KOH), Thionyl chloride ($SOCl_2$), triethylamine (TEA), N, N-dimethylformamide (DMF), Acetic acid (CH_3COOH) and Ammonia (NH_3) analytical grade reagents and chemicals were obtained from Kobian Kenya Limited, which is a Sigma Aldrich's outlet in Kenya. Distilled water was used to prepare reagents and for dilution.

2.3 Preparation of fluoride stock solution

A fluoride standard stock solution (1000 mg/L) was prepared by dissolving 2.21 g of NaF salt in a total ionic strength adjustment buffer (TISAB) media in a 1000 mL mark in a volumetric flask to maintain a constant ionic strength. Then, the prepared solution was used to prepare subsequent dilute working solutions. A solution of 1.0×10^{-1} mol dm^{-3} KOH and HCl was used to adjust the working solutions' pH to the values dictated by each experiment. Fresh dilutions were used in each adsorption experiment.

The total ionic strength adjustment buffer (TISAB) was prepared by mixing 57 ml of acetic acid, 58 g of sodium chloride and 4 g of 1,2-cyclohexanedinitrilo-tetraacetic acid (CDTA) in 1000 ml of distilled water.

2.4 Sample collection and pre-treatment

The watermelon rind biowastes were randomly collected around Othaya markets in Nyeri County, Kenya (−0.551751'S, 36.944703'E). They were then transported to the laboratory, cleaned thoroughly with distilled water, chopped into small pieces, oven-dried for 1 day at 105 °C to remove moisture, ground into a fine powder and kept in an airtight bottle awaiting subsequent experiments.

2.5 Chemical modification

Chemical modification was done by first chlorination followed by amination of watermelon rind biowastes as described by [13]. This involved grinding the dried sample, followed by pre-activation at 80 °C for 12 h. The pre-activated powder (10.000 g) was

transferred into a three necked flask containing DMF (200 mL). This was followed by slowly adding $SOCl_2$ (35 mL) and mechanically stirring the mixture under reflux for 4 h at 80 °C. This was followed by washing the chlorinated material thrice with 2 M NH_4OH (25 mL) solution to adjust the pH to neutral and finally with double-distilled water. The material was filtered and vacuo dried at a room temperature. The product (5.000 g) was transferred to a three necked flask that containing DMF (100 mL). 25 mL of the TEA was added slowly while stirring under reflux continued for 3 h at 80 °C. The resultant mixture was filtered and the modified adsorbent vacuo dried at a room temperature and stored in airtight container awaiting characterization and subsequent studies.

2.6 Instruments and equipments

The used instruments and equipment were Field Emission Scanning Electron Microscope (FEI ESEM, Vega3 Tescan LMH), pH meter (PHS-3 C), X-Ray Diffractometer, XRD (Rigaku MiniFlex II; made in Tokyo, Japan), Grinding mill (Retsch SR 200), Thermogravimetric Analyzer (TGA-50 SHIMADZU), Drying oven (WTC binder FD53), Distiller (WSB 14), Fourier Transform Infra-red (FT-IR) spectrophotometer (JASCO FT/IR-4700), Lab-line mechanical reciprocating shaker model (DKZ-1NO.1007827), Analytical weighing balance (ATX224 Shimadzu) and Fluoride ion selective electrode (HACH ISEF12101).

2.7 Batch experiments

Adsorption experiments were performed at a room temperature (298 K) by batch mode on a mechanical shaker at 150 rpm using plastic screw cap bottles (100 mL). The process parameters of pH (1–13), contact time (10–130 min), adsorbent mass (10–130 mg) and initial concentration (50–170 mg/L) were varied in 40 mL of the test solution. This was done by adjusting a single parameter while keeping the other parameters constant (OVAT). The experiments were done thrice at an agitation speed of 150 rpm. The amount of fluoride ions adsorbed per unit mass of the adsorbent was determined using Eq. 1.

$$\%\,\mathrm{Removal} = \frac{(\mathrm{C_i} - \mathrm{C_e})}{\mathrm{C_i}} \times 100 \tag{1}$$

Where, C_i is the initial fluoride ion concentration (mg/L) and C_e is the fluoride ion concentration at equilibrium (mg/L).

2.8 Experimental design and statistical analysis

Central composite design (CCD) was applied as an optimization tool of RSM, to explore the impact the interactions of the various parameters has on the sorption process. To determine the optimum conditions, four process parameters were considered: pH (X_1), contact time (X_2), adsorbent mass (X_3), initial concentration (X_4) and their interactive influence on the F^- ions uptake. The variables and their levels are presented in Table 1.

The CCD involves 2^n factorial runs, 2n axial runs and n_c center runs. The total experimental runs conducted are computed by Eq.

$$\mathrm{N} = 2^{\mathrm{n}} + 2\mathrm{n} + \mathrm{n_c} \tag{2}$$

Table 1 Experimental ranges and levels of independent factors

Factors	Factors code	Central composite levels		
		Low (−1)	Medium (0)	High (+1)
pH	X_1	1	7	13
Contact time (min)	X_2	10	70	130
Adsorbent mass (mg)	X_3	10	70	130
Initial concentration (mg/L)	X_4	50	110	170

Table 2 Experimental runs generated by CCD in minitab software

Run	Actual values			
	pH (X_1)	Contact time (X_2, min)	Adsorbent mass (X_3, mg)	Initial concentration (X_4, mg/L)
1	7	70	70	230
2	13	130	10	170
3	1	70	70	110
4	7	70	70	110
5	19	70	70	110
6	13	10	130	50
7	7	70	70	110
8	7	70	50	110
9	13	10	130	170
10	1	10	130	170
11	7	70	190	110
12	7	70	70	110
13	7	70	70	10
14	1	10	10	50
15	1	130	10	50
16	1	50	70	50
17	13	130	130	170
18	7	70	70	110
19	7	70	70	110
20	1	10	10	170
21	13	130	130	50
22	7	190	70	110
23	1	10	130	50
24	1	130	10	170
25	7	70	70	110
26	13	10	10	50
27	1	130	130	170
28	13	130	10	50
29	1	130	130	50
30	7	70	70	110
31	13	10	10	170
32	7	50	50	50

where n is number of independent variables (factors), nc is number of center points and N is the overall total of experimental runs. This indicated that 32 experimental runs comprising 16 factorial runs, 8 axial runs and 8 center runs were prerequisites for the modeling and optimization process. This is as shown by Table 2.

The optimum values of the test variables were obtained using the numerical point prediction tool in MINITAB statistical software (Version 17.0, PA, USA). The optimum values of the process variables were first obtained in coded units and then converted to uncoded forms.

2.9 Adsorption isotherms

The experimental data obtained at optimal process parameters was modelled using Langmuir and Freundlich isotherm to determine the maximum adsorption capacity of the modified adsorbent. The linearized equation for Langmuir isotherm is expressed in Eq. 2.

$$\frac{C_e}{q_e}=\frac{C_e}{q_m}+\frac{1}{k_L q_m} \tag{3}$$

Where q_e and q_m is the respective experimental and calculated sorption capacity (mg/g), K_L (L/g): the constant and C_e (mg/L): the concentration at equilibrium.

The equation for Freundlich isotherm is given by Eq. 3.

$$\ln q_e=\ln K_F+\frac{1}{n}\ln C_e \tag{4}$$

Where $\frac{1}{n}$ (adsorbent heterogeneity), K_F (sorption capacity, mg/g) and n (adsorption intensity) parameters are calculated from linear plots of $\ln q_e$ against $\ln C_e$.

2.10 Kinetic studies

The time data was fitted using pseudo-first-order and pseudo second-order kinetic models to determine the rate and adsorption mechanisms. The linearized kinetic rate equation for pseudo-first-order and pseudo second-order model can be written a given by Eqs. 4 and 5.

$$\ln\left(q_e-q_t\right)=\ln q_e-K_1 t \tag{5}$$

$$\frac{t}{q_t}=\frac{1}{k_2 q_e^2}+\frac{1}{q_e}t \tag{6}$$

Where q_e (mg/g) and q_t (mg/g): the respective uptake capacity at equilibrium and at time (t) while K_1 (min^{-1}) and k_2 (mg/g/min) are rate constants.

2.11 Adsorbent performance using environmental water samples

The environmental water samples were obtained from Lake Baringo. They were taken randomly from seven different sampling points along the shores of the river. Water samples of 1 L were taken from each sampling point along the randomly selected 7 sampling points on the shores of Lake Baringo to provide a wide range in water composition and concentrated HNO_3 acid (5 mL) solution added to each water sample, bottles sealed and samples were transported to the laboratory. A 50 mL of the water sample from each of the sampling point was placed in a beaker. The analysis of the amount of F^- ions in the solution found in environmental water samples was determined. Then, batch experiments using the modified adsorbent were executed at optimal pH (1), contact time (50 min) and adsorbent mass (70 mg) and the amount of fluoride ions determined. The retained fluoride ions in the adsorbent were stripped with 1.0×10^{-1} mol dm^{-3} NaOH eluent solution. The regeneration efficiency was calculated using Eq. 6.

$$\text{Regeneration efficiency }(\%)=\frac{c_{des}}{c_{ads}}\times 100 \tag{7}$$

Where, C_{ads} and C_{des} (mg/L) are the respective flouride concentration adsorbed and desorbed.

3 Results and discussion

3.1 Adsorption mechanism

The mechanism of adsorption formed at the surface of the adsorbent with fluoride ions is shown by Fig. 2.

The modified adsorbent provides many centers for fluoride ions adsorption on their surfaces compared to the raw biomass material. The main mechanism of uptake is mainly electrostatic interactions (ion exchange) between the negatively charged fluoride (F^-) ions and the positively charged amine groups ($^+N(CH_2CH_3)_3$) on the adsorbent surface. The presence of more binding centers could improve its adsorption properties in defluoridation of water.

3.2 Characterization

The raw biomass and the modified adsorbent from watermelon rind biowastes was characterized using FT-IR, TGA and SEM-EDX. The results are discussed in the following sub-sections.

3.2.1 FT-IR characterization

The FT-IR analysis for the raw biomass and modified adsorbent was carried out at a mid-IR range of 4000–500 cm^{-1} and at average of 10 scans per spectrum and a resolution of 4 cm^{-1}. The results are shown by Fig. 3.

The results in FT-IR spectrum of raw biomass showed a broad band at 3219.58 cm^{-1} is ascribed to hydroxyl groups (-OH) stretching vibrations of alcoholic and carboxylic acid groups [19]. A weak peak at 2914.92 cm^{-1} is attributed to symmetric and asymmetric sp^3 –CH stretching vibrations [20]. The absorbance at 1602.56 cm^{-1} correspond to carbonyl (C = O) stretch in carboxylic acid and its derivatives [21]. An absorbance at 1511.69 cm^{-1} could be due to carbon-carbon (C = C) stretching vibrations of aromatic hydrocarbons [22]. This The peaks at 1292.64 cm^{-1} and 1089.88 cm^{-1} could be due to C-O stretch vibrations of carboxylic acid derivatives and alcoholic groups [23].

The results in FT-IR spectrum of modified adsorbent shows that the (-OH) absorption peak is shifted to 3359.32 cm^{-1} from 3219.58 cm^{-1} (raw) an indication that some –OH groups were replaced with $–NR_3$ groups during the modification process. The shifting of symmetric $–CH_2$ stretching vibration from 2914.92 cm^{-1} (raw) to 2913.64

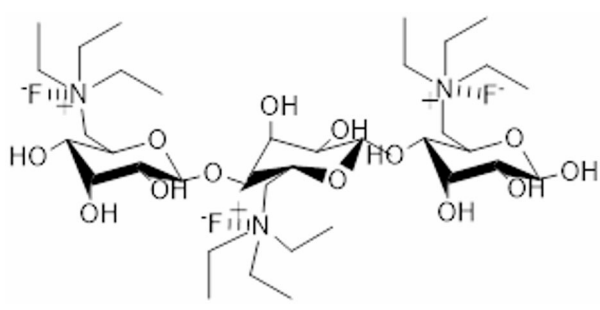

Fig. 2 Adsorption mechanism of F^- ions onto the modified adsorbent

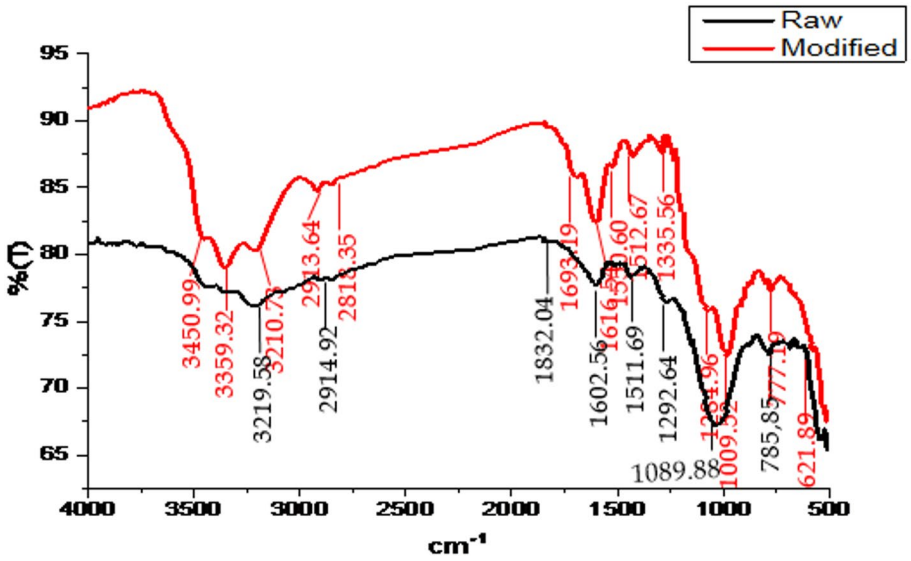

Fig. 3 FT-IR spectrum of raw biomass and the modified adsorbent

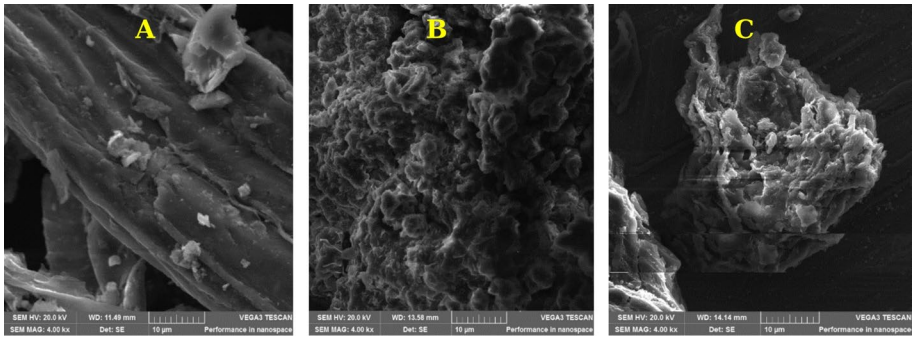

Fig. 4 SEM images of raw biomass (**A**), Modified adsorbent (**B**) and fluoride loaded adsorbent (**C**)

cm^{-1} (modified) together with a new band at 2818.35 cm^{-1} is an indication of methylene groups of TEA moiety in the modified adsorbent [24]. The band at 621.89 cm^{-1} could be assigned to –CN stretching [14]. The results showed a successful synthesis of the adsorbents.

3.2.2 SEM characterization

The SEM micrograph for the raw biomass and the modified adsorbent were obtained at 20.0 kV (voltage) and 6.50 K X (magnification). The results are as shown in Fig. 4A, B and C.

From the results in Fig. 4, the SEM micrographs showed that the surface texture and morphology of the raw biomass material was rigid, regular, less porous and constrained, as seen in Fig. 4A. After chemical modification, the adsorbent surface became rugged and more porous (Fig. 4B). This was attributed to cellulose and hemicellulose breakdown due to heat during the chemical modification [23].

Before adsorption of F^- ions (B), the adsorbent surface evidenced pores of different shapes and sizes. The fluoride-loaded adsorbent surface (C) is more rigid and restrained which is contributed to fluoride ions having occupied the pores. The results showed that the porous nature of the adsorbent also facilitated the uptake process.

3.2.3 EDX characterization

The elemental constitution of the raw biomass and the modified adsorbent was determined and the results are as shown by the Figs. 5A and B.

From the EDX results in Fig. 5, the main elemental composition of the raw biomass was carbon (60.1%), oxygen (36.1%) and potassium (3.8%). This showed that carbon (C) was the main constituent of raw biomass material. The elemental oxygen could also be as attributed to surface oxides of potassium element.

The presence of nitrogen content (11.5%) in the modified adsorbent showed that TEA was successively anchored on the surface of the watermelon rind biowastes. The percentage of carbon decreased from 60.1% (raw) to 52.7% (modified). This is due to the degradation of cellulose polymeric compounds during the pre-activation, chlorination and amination processes [24].

3.2.4 TGA characterization

The evaluation of thermal stability of the raw biomass and the modified adsorbent was done at a varied temperature of 0 °C to 600 °C. The TG curves are presented by Fig. 6.

In a temperature range of 10–100 °C, the raw biomass material experienced a weight loss from 8.8 g to 6.6 g (25%) due to the vaporization of free water, hydrocarbons and other volatile organic compounds [25]. In contrast, the modified adsorbent exhibited a lower weight loss range (9.0 g–7.0 g) of around 22.22%. The modified adsorbents exhibited a maximum weight loss temperature range between 100 and 400 °C, demonstrating that it is more thermally stable even at high temperatures and therefore suitable in treatment of fluoride laden wastewater.

3.3 Batch optimization studies

The impact of adsorbent mass, contact time, pH and initial concentration on F^- ions uptake onto the modified adsorbent were investigated. The chosen process parameters are the most key and impactful in an uptake process. The results are presented by Fig. 7A, B, C and D.

From the results in Fig. 7A, a maximum removal efficiency of 60.09% was obtained at an optimal pH of 1. This means that the uptake of F- ions onto the modified adsorbent is best in strongly acidic media. This followed a decrease in uptake to 31.94% with increased in solution pH. At a pH of 1, the H^+ ions concentration is raised initiating

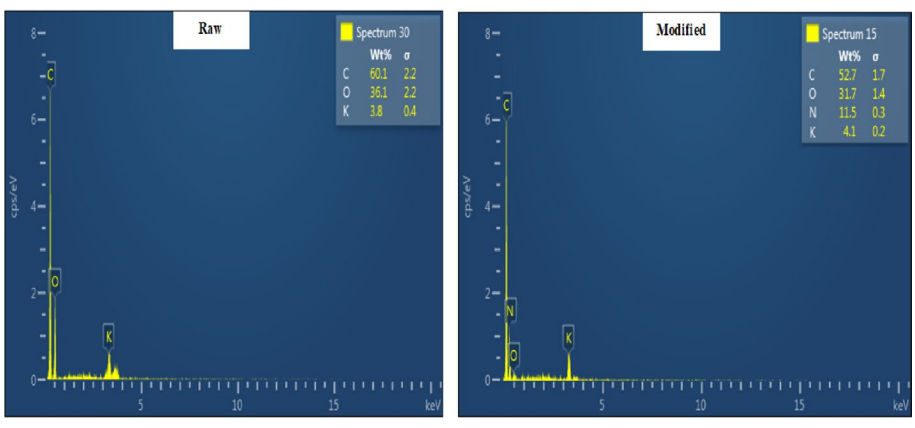

Fig. 5 The EDX spectra of raw biomass and the modified adsorbent

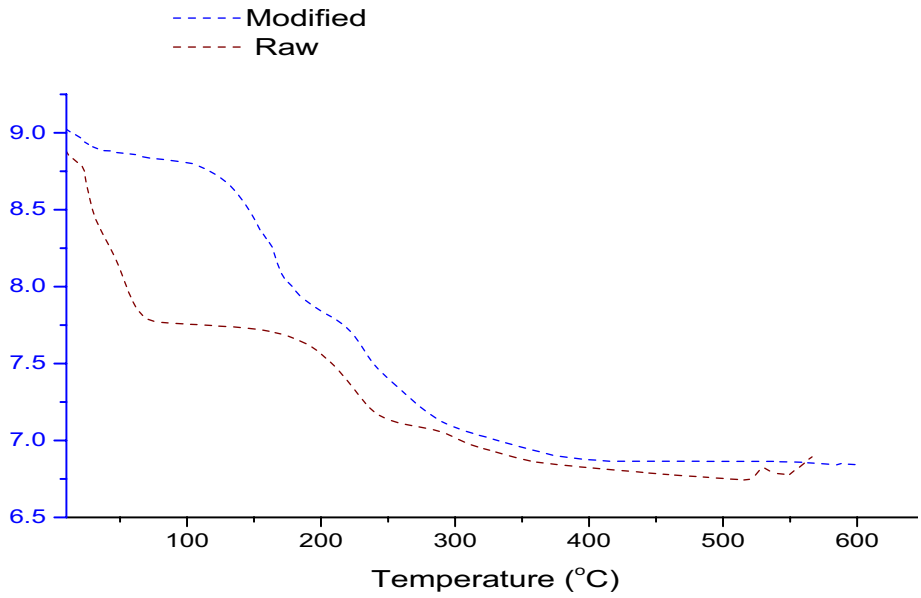

Fig. 6 The TG curve for raw biomass and the modified adsorbent

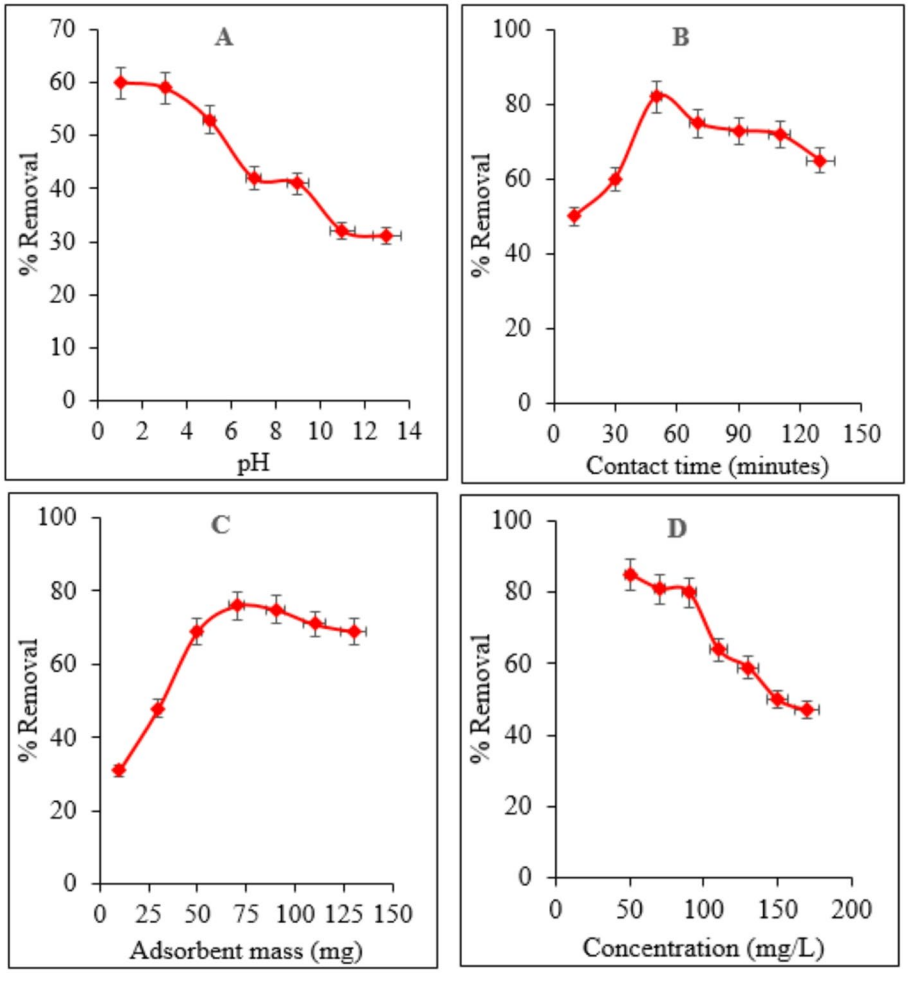

Fig. 7 Effect of pH (**A**), contact time (**B**), adsorbent mass (**C**) and initial concentration (**D**) on F^- ions uptake

electrostatic repulsions between the ions in the solution and the positively charged amine groups in the adsorbent. This favour adsorption of F^- ions. At pH > 1, the solution becomes more alkaline as a result of excess OH^- ions causing its competition with the F_- ions for the adsorbent sites weakening the efficiency of the modified adsorbent [26]. Similar observations have been reported by [27].

According to the results in Fig. 7B, increasing contact time of interaction increases the F^- ions from 51.39% to maximum of 82.43%, at optimal value (50 min) beyond which no significant change was registered. The rapid uptake of F^- ions at initial stages was due to the availability of binding sites leading to faster rate of adsorption [28]. Subsequently, the uptake decreased beyond 50 min due to the saturation of all the active sites present on the adsorbent which do not allow for further adsorption to take place [1]. These findings compare well with those reported by [29].

The results in Fig. 7C show that the percentage of F^- ions removal increased from 31.67 to 76.11% with an increase in adsorbent mass from 10 mg to 70 mg beyond which an insignificant decrease is observed. The results illustrate an increase in removal uptake with an increase in adsorbent mass, because of an increase in the number of available active sites on the adsorbent surface until the adsorption achieved equilibrium [2]. Similar trend is reported in the literature [30].

The results (Fig. 7D) showed that the removal efficiency decreased from 85.97% to 47.64% as the concentration was increased from of 50 mg/L to 170 mg/L. This can be attributed to availability of a large number of adsorption sites at low concentration which gets saturated as the concentration increases.

3.4 Multivariate (statistical) analysis

3.4.1 Regression model equation

In the present study, four process parameters, pH (X_1), contact time (X_2), adsorbent mass (X_3) and initial concentration (X_4) were chosen as the independent variables while the removal of F^- ions at equilibrium (Y) was selected as the response (dependent variable) of the study. The final central composite design obtained for percentage removal with significant terms was quadratic as suggested by the software, and is given by Eq. 6.

$$\begin{aligned} Y = {} & 65.89 + 0.705\ X_1 - 0.0000\ X_2 - 0.0172\ X_3 \\ & + 0.1459\ X_4 - 0.0228\ (X_1)^2 + 0.000451\ (X_2)^2 \\ & + 0.000233\ (X_3)^2 - 0.000300\ (X_4)^2 - 0.00404\ X_1X_2 \\ & + 0.00486\ X_1X_3 - 0.00519\ X_1X_4 + 0.000356\ X_2X_3 \\ & - 0.000725\ X_2X_4 - 0.000286\ X_3X_4 \end{aligned}$$

The predicted removal efficiency (%) obtained by Eq. 6 shows experimental design matrix with response is shown in Table 3.

The residuals between the predicted and experimental (actual) values were found to be low implying that both the predicted and experimental responses had strong agreements. This was confirmed by a parity linear plots as shown by the Fig. 8.

The correlation coefficient (R^2) used to determine the relationship between the experimental (actual) and predicted responses was 0.9256 as established by the model. This suggests that the process parameters analyzed explain 92.56% of the % removal variability and that the model could not explain only 7.44% of the variation of the response. The derived quadratic equations using the CCD matrix were found to be suitable for

Table 3 Experimental design matrix with response

Run	Actual values				Coded values				Removal efficiency (%)		
	pH	Contact time	Adsorbent mass	Initial concentration	X_1	X_2	X_3	X_4	Experimental	Predicted	Residual
1	7	70	70	230	0	0	0	2	68.25	67.06	1.19
2	13	130	10	170	1	1	−1	1	60.64	61.10	−0.46
3	1	70	70	110	−2	0	0	0	72.59	74.59	−2.00
4	7	70	70	110	0	0	0	0	73.55	74.64	−1.09
5	19	70	70	110	2	0	0	0	70.59	69.82	0.77
6	13	10	130	50	1	−1	1	−1	82.54	82.05	0.49
7	7	70	70	110	0	0	0	0	75.11	74.64	0.47
8	7	70	50	110	0	0	−2	0	72.01	73.88	−1.86
9	13	10	130	170	1	−1	1	1	77.93	78.21	−0.28
10	1	10	130	170	−1	−1	1	1	78.18	77.07	1.11
11	7	70	190	110	0	0	2	0	82.45	83.14	−0.69
12	7	70	70	110	0	0	0	0	70.32	74.63	−4.32
13	7	70	70	10	0	0	0	−2	75.18	74.36	0.82
14	1	10	10	50	−1	−1	−1	−1	70.02	72.29	−2.27
15	1	130	10	50	−1	1	−1	−1	76.98	75.46	1.52
16	1	50	70	50	0	−2	0	0	73.23	72.49	0.74
17	13	130	130	170	1	1	1	1	69.78	70.25	−0.47
18	7	70	70	110	0	0	0	0	75.43	74.64	0.79
19	7	70	70	110	0	0	0	0	74.41	74.64	−0.23
20	1	10	10	170	−1	−1	−1	0	79.12	80.04	−0.92
21	13	130	130	50	1	1	1	−1	83.69	84.53	−0.84
22	7	190	70	110	0	2	0	0	77.48	78.74	−1.26
23	1	10	130	50	−1	−1	1	−1	72.13	73.44	−1.31
24	1	130	10	170	−1	1	−1	1	74.56	72.77	1.79
25	7	70	70	110	0	0	0	0	76.59	74.64	1.95
26	13	10	10	50	1	−1	−1	−1	74.93	73.91	1.02
27	1	130	130	170	−1	1	1	1	75.14	74.93	0.21
28	13	130	10	50	1	1	−1	−1	72.42	71.26	1.16
29	1	130	130	50	−1	1	1	−1	82.69	81.73	0.96
30	7	70	70	110	0	0	0	0	74.46	74.64	−0.18
31	13	10	10	170	1	−1	−1	1	75.55	74.18	1.32
32	7	50	50	50	1	0	−1	0	73.69	73.94	−0.25

predicting independent process variables' influence on removal efficiency of for F^- ions onto the modified adsorbent [31]. The results are in good coherent with those reported by [32].

The adequacy of the models was justified by the analysis of variance (ANOVA). The ANOVA of fluoride ions adsorption is given in Table 4.

The model F-value for percentage removal of 22.55 implies the model is significant. Also, low p-values for removal efficiency less than 0.05 except for $(X_4)^2$ indicate model terms are significant. The Lack-of-Fit F-value for % removal of 0.23 implies the Lack of-Fit is significant. High F-value, correlation coefficient (R) tending to 1, low p-value and most importantly the results of lack of fill test indicates the high statistical significance of the developed model with high accuracy [33]. The removal of the F^- ions is influenced significantly by the interaction of contact time and concentration with F value of 66.41 and less with other parameters interactions 35.48, 16.23, 36.77 and 7.37 for pH and contact time, pH and adsorbent mass, pH and concentration, time, adsorbent mass and concentration respectively.

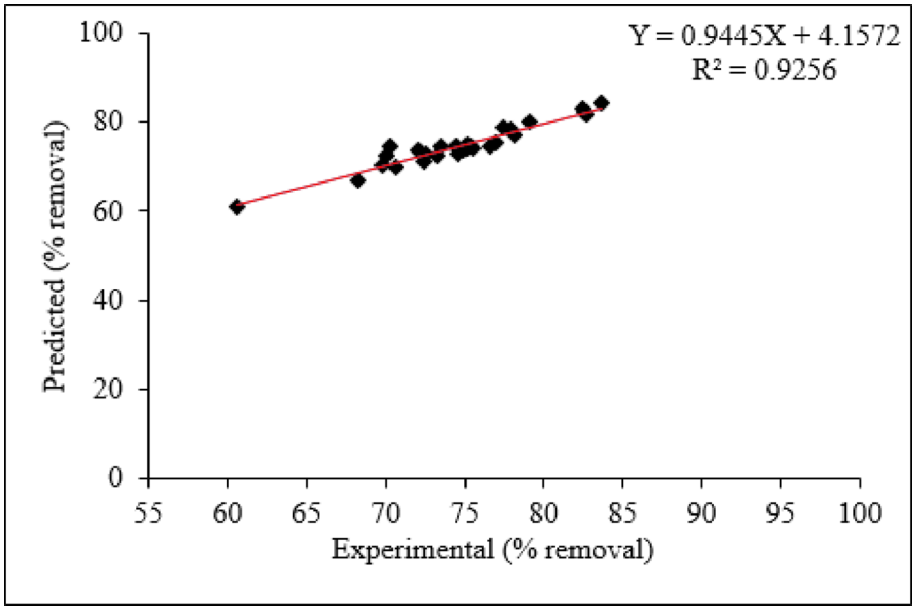

Fig. 8 Parity plot for the experimental and predicted % removal value for F^- ions uptake

Table 4 ANOVA analysis for CCD response surface quadratic model for fluoride ions adsorption

Response source	Degree of freedom (DF)	Sum of squares (SS)	Mean square (MS)	F value	*P* value	Comment
Model	14	642.473	45.891	22.55	0.000	Significant
Linear	4	186.526	46.631	22.91	0.000	Significant
X_1	1	11.844	11.844	5.82	0.028	Significant
X_2	1	15.626	15.626	7.68	0.014	Significant
X_3	1	96.330	96.330	47.33	0.000	Significant
X_4	1	62.711	62.711	30.81	0.000	Significant
Quadratic	4	66.366	16.592	8.15	0.001	Significant
$(X_1)^2$	1	9.268	9.268	4.55	0.049	Significant
$(X_2)^2$	1	29.840	29.840	14.66	0.001	Significant
$(X_3)^2$	1	13.731	13.731	6.75	0.019	Significant
$(X_4)^2$	1	6.530	6.530	3.21	0.092	Not significant
2-way interaction	6	352.510	58.752	28.87	0.000	Significant
$X_1 \times_2$	1	72.204	72.204	35.48	0.000	Significant
$X_1 \times_3$	1	33.034	33.034	16.23	0.001	Significant
$X_1 \times_4$	1	74.825	74.825	36.77	0.000	Significant
$X_2 \times_3$	1	14.996	14.996	7.37	0.015	Significant
$X_2 \times_4$	1	135.164	135.164	66.41	0.000	Significant
$X_3 \times_4$	1	14.996	14.996	7.37	0.015	Significant
Residual error	16	32.563	2.035			
Lack of fit	10	8.953	0.895	0.23	0.980	Not significant
Pure error	6	23.610	3.935			
Total	30	675.036				

The results compiled by ANOVA predict the adequate representation of the actual relation between the response function, i.e., removal %, and independent variables (pH, adsorbent dose, initial concentration and contact time) by a second-order polynomial equation. The results mirror those reported in the literature [34–36].

3.4.2 Response surface plots

The 3D surface plots indicating the combined effects of pH (X_1), contact time (X_2), adsorbent mass (X_3) and initial concentration (X_4) and their effect on the removal efficiency (%) of F^- ions are shown in Fig. 9A, B, C, D, E and F.

The 3D surface plots of the interactive effect of pH, contact time, adsorbent mass and initial concentration on the adsorption of F- ions, are presented in Figs. 10A–F. These 3D plots are analysed to pinpoint the optimal or near-optimal adsorption areas. The adsorption process is greatly influenced by the interaction of contact time and concentration followed by that of pH and concentration. The interaction of parameters confirmed what is reported by Table 4. From the surface plots, the uptake of F^- ions by the modified adsorbent is predicted by CCD RSM to be high at lower pH values (1), contact time (50 min), higher adsorbent mass (70 mg) and less concentrations (50 mg/L). The results confirm optimal conditions reported earlier in Fig. 7A–D (OVAT). Similar findings are reported by [37].

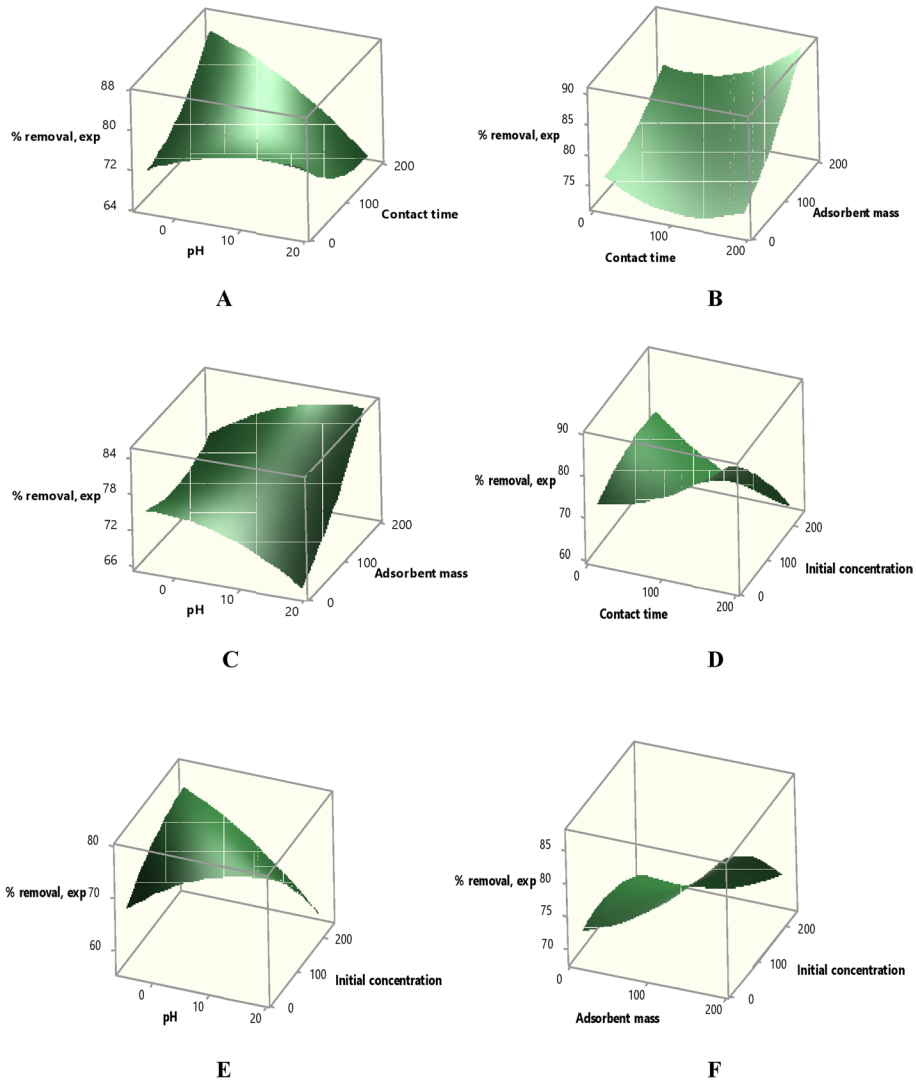

Fig. 9 **A**, **B**, **C**, **D**, **E and F**: 3D surface plots showing interactive effects of chosen variables in F^- ions adsorption

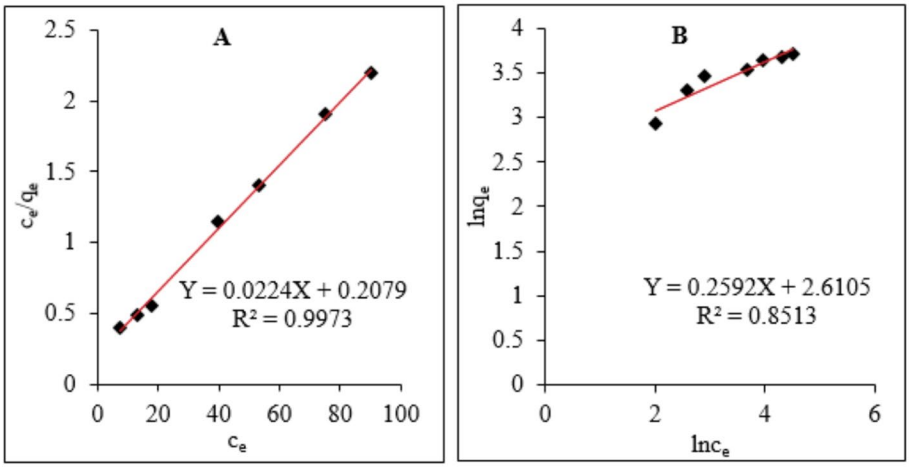

Fig. 10 Langmuir (**A**) and Freundlich (**B**) Isotherm plots

Table 5 Langmuir and Freundlich parameters for F^- ions adsorption onto the modified adsorbent

Langmuir				Freundlich			Best model
Q_{exp} (mg/g)	Q_{cal} (mg/g)	K_L (L/g)	R^2	1/n	K_F (mg/g)	R^2	
41.09 ± 0.09	44.64 ± 0.02	1.08×10^{-1}	0.9973	2.59×10^{-1}	13.61	0.8513	Langmuir

Table 6 Adsorption capacity of different adsorbents for the adsorption of fluoride ions

Adsorbent	Adsorption capacity (mg/g)	Reference
MgO nanocomposites	15.59	[40]
Fe(III)-doped scoria	0.32	[41]
carbon@SnO_2 nano composites	4.89	[42]
Silicon nitride	13.90	[29]
Bio- filters sorbent	5.53	[43]
Polythene	18.55	[15]
Construction wastes	15.16	[44]
Metallic phosphates	31.61	[45]
Activated carbon supported Ce–Al oxides	17.97	[46]
Modified watermelon rind	41.09	Current study

3.5 Efficacy studies

The plotting of $\frac{C_e}{q_e}$ against C_e (Langmuir) and $\ln q_e$ against $\ln C_e$ (Freundlich) yields graphs were employed to determine isotherm parameters and constants. The results are represented in Table 5.

From the outcomes in Table 5, correlation coefficients (R^2) of > 0.99 were obtained for Langmuir isotherm. This was corroborated by the closeness of Q_{exp} (mg/g) and Q_{cal} (mg/g). This implies that the Langmuir gave better regression coefficient than those of Freundlich. This is as shown by Fig. 10. This suggested a monolayer chemisorption interaction between the F^- ions and the adsorbent sites [38]. The values of $K_L < 1$ showed the favorability of F^- ions sorption onto the adsorbent [39]. The results mirror those reported by [16].

A comparable adsorption capacity of the adsorbent under study with other adsorbents for F- ions removals is shown in Table 6.

Table 7 Pseudo-first-order and Pseudo-second-order parameters for F- ions adsorption

Pseudo-first-order				Pseudo-second-order			Best model
Q_{exp} (mg/g)	Q_{cal} (mg/g)	K_1 (min^{-1})	R^2	Q_{cal} (mg/g)	K_2 (mg $g^{-1}min^{-1}$)	R^2	
19.56 ± 0.02	6.02 ± 0.01	6.30×10^{-3}	0.2100	17.89	5.56×10^{-3}	0.9486	Pseudo-second-order

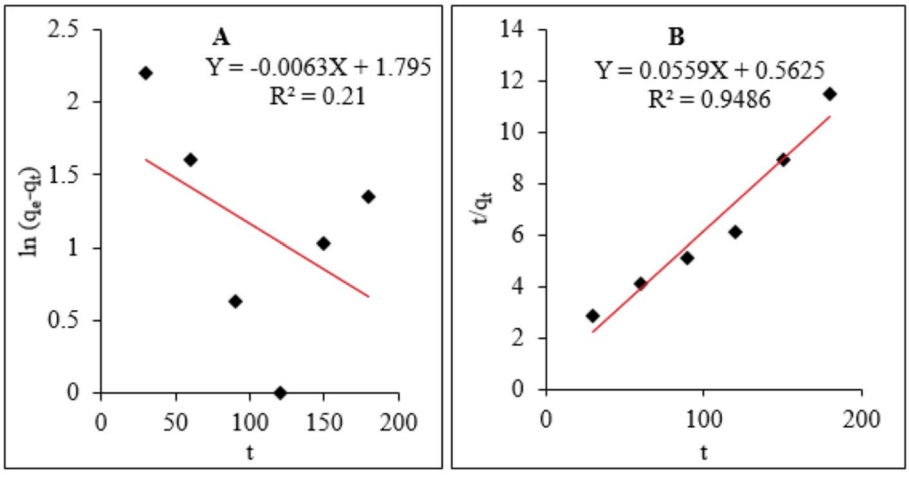

Fig. 11 Pseudo-first-order (**A**) and Pseudo-second-order (**B**) kinetic plots

Table 8 Analysis of environmental water samples

Found in environmental water sample (mg/L)	Found in water sample after adsorption (mg/L)	(% Removal ± S.D)
3.59 ± 0.03	0.76 ± 0.02	78.83 ± 0.06

3.6 Kinetic studies

The two different kinetic parameters (Pseudo-first-order and Pseudo-second-order) calculated by linear regression of time data for F^- ions uptake are shown in Table 7.

Pseudo-second-order kinetic plots showed a better correlation with the experimental data than Pseudo-first-order plots as shown by the Fig. 11. Kinetic data shows R^2 value of 0.2100 for pseudo-first-order model and 0.9486 for pseudo-second-order model. The experimental Q_{exp} value (19.56 mg/g) is closer to calculated Q_{cal} value (17.89 mg/g) as compared to those in pseudo-first-order (6.02 mg/g). The K_2 value (5.56×10^{-3} (mg $g^{-1}min^{-1}$) is higher than K_1 value (6.30×10^{-3} (min^{-1}).

The closeness of Q_{cal} and Q_{exp} with higher R^2 values (> 0.94) and higher K_2 values indicated that the pseudo-second-order model best fit the experimental data for F^- ions removal compared to pseudo-first-order. The model assumes chemisorption as the rate determining step that controlled the adsorption process [47]. Mutai et al. [11] made similar observations.

3.7 Analysis of environmental water samples

Analysis of fluoride levels in the water sample from Lake Baringo and the batch results are presented in Table 8.

Table 9 Regeneration efficiency of the modified adsorbent in F^- ions adsorption

Cycle	Found (mg/L)	Desorbed (mg/L)	Adsorbed (mg/L)	Regeneration efficiency (% *R* ± S.D)
1	3.59 ± 0.03	2.01 ± 0.01	2.83 ± 0.00	71.02
2	3.59 ± 0.03	0.64 ± 0.02	1.05 ± 0.02	60.95
3	3.59 ± 0.03	0.44 ± 0.00	0.81 ± 0.01	54.32
4	3.59 ± 0.03	0.29 ± 0.00	0.56 ± 0.00	51.79

The results in Table 8 showed that the level of fluoride ions in Lake Baringo water was 3.59 mg/L higher than the recommended WHO limits (1.5 mg/L). The pH of the water sample (not reported in the table) was 7.50. After treatment using the modified adsorbent, the concentration reduced to 0.76 mg/L with removal efficiency of 78.83%. The lower efficiency than the reported using the synthetic solution (85.97%) could be due to high pH values (7.50) and the existence of other interfering species in the water samples such as NO_3^-, Cl^- and NO_2^-which competed for the available adsorbent sites with the F^- ions resulting to faster saturation of the adsorbent sites [48, 49].

3.8 Regeneration studies

The reusability of the modified adsorbent in F^- ions adsorption was investigated, and the results are presented in Table 9.

From the findings in Table 9, regeneration experiments revealed that desorption percentage decreased as the adsorption/desorption cycles advanced upto 4th cycle. The decrease in the uptake capacity was due to the ignorable mass of the adsorbent that got lost during the adsorption-desorption processes [50]. The regeneration efficiency was > 50%. The results demonstrated that the modified adsorbent can be re-used repeatedly for defluoridation of water with a high efficiency lowering the fluoride levels to (< 1.5 mg/L (below WHO permissible levels).

4 Conclusions

In the present work, raw watermelon rind biowastes were chemically modified with thionyl chloride followed by triethylamine in a N, N-dimethylformamide solvent. This was confirmed by FT-IR, SEM-EDX and TGA results which showed a thermally stable and porous adsorbent surface with the presence of functional groups of carbonyl (-C = O), hydroxyl (OH-), carboxylate (-COO-) and additional anchored amine ($-^+N(CH_2CH_3)_3$) groups (modified adsorbent) as important adsorption sites for fluoride (F^-) ions removal. The adsorption process was greatly influenced by adsorption parameters of pH, contact time, adsorbent mass and initial concentration. The optimum values were pH (1), 50 min, 70 mg and 50 mg/L. The central composite design (CCD) response surface plots confirmed the optimal batch optimization experiments. The uptake of fluoride (F^-) ions was well described by Langmuir isotherm model with a maximum adsorption capacity of 44.64 ± 0.02 mg/g. This implied a chemical interaction between fluoride (F^-) ions and the adsorbent sites. Pseudo-second-order kinetic model fitted well in describing adsorption of F^- ions and that chemisorption was the rate determining step. A regeneration efficiency of > 50% was obtained for the adsorption/desorption cycles for F^- ions uptake.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s44371-025-00449-9.

Supplementary material 1

Author contributions
Author 1 and 3 (Samuel Ndung'u & Thomas Nyahanga) contributed in the concept and design of the research study, data collection and interpretation of data. Author 2 (Esther Kinuthia) contributed to drafting of the manuscript. Author 4 (Samuel Kirkok) contributed to critical review and final approval for submission to Discover Chemistry Journal.

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Data availability
All the data obtained as results of the research study are available in this manuscript. There is no hidden or unavailable data.

Declarations

Ethics approval and consent to participate
This declaration is not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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